



AWS VPC

Virtual Private Cloud

AWS VPC Essentials

Isolated Cloud Networks

A VPC allows you to launch resources in a logically isolated virtual network, enhancing privacy within the cloud.

Customizable Network Settings

Users can define IP address ranges, create subnets, and set up route tables for flexible network architecture.

Enhanced Security Controls

VPC provides security groups and network ACLs to control inbound and outbound traffic, protecting cloud resources.



AWS VPC Key Components

Subnets and Route Tables

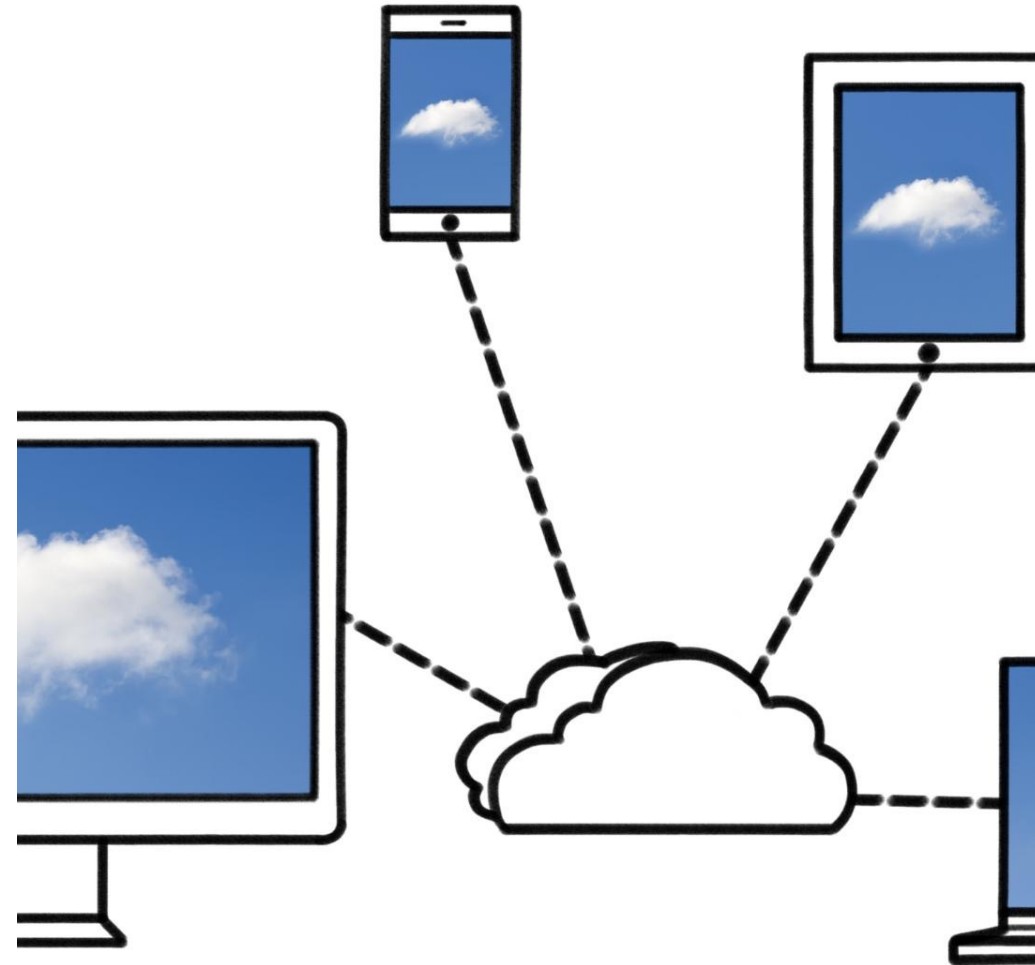
Subnets organize resources by dividing the IP range. Route tables manage network traffic flow between these subnets and other networks.

Gateways for Connectivity

Internet Gateways provide external connectivity, while NAT Gateways enable private subnets to access the internet securely.

Network Security Controls

Security Groups serve as virtual firewalls, and Network ACLs offer stateless traffic filtering for added protection.





Why AWS VPC Matters

Secure Cloud Environments

Amazon VPC creates isolated environments, protecting cloud resources from unauthorized access and threats.

Granular Network Control

VPC enables detailed management of network access, allowing precise control of data traffic within the cloud.

Customizable Connectivity

Organizations can customize network configuration to optimize links between on-premises and cloud infrastructure, ensuring compliance.

AWS VPC IPv4 CIDR Range

Defining IP Address Space

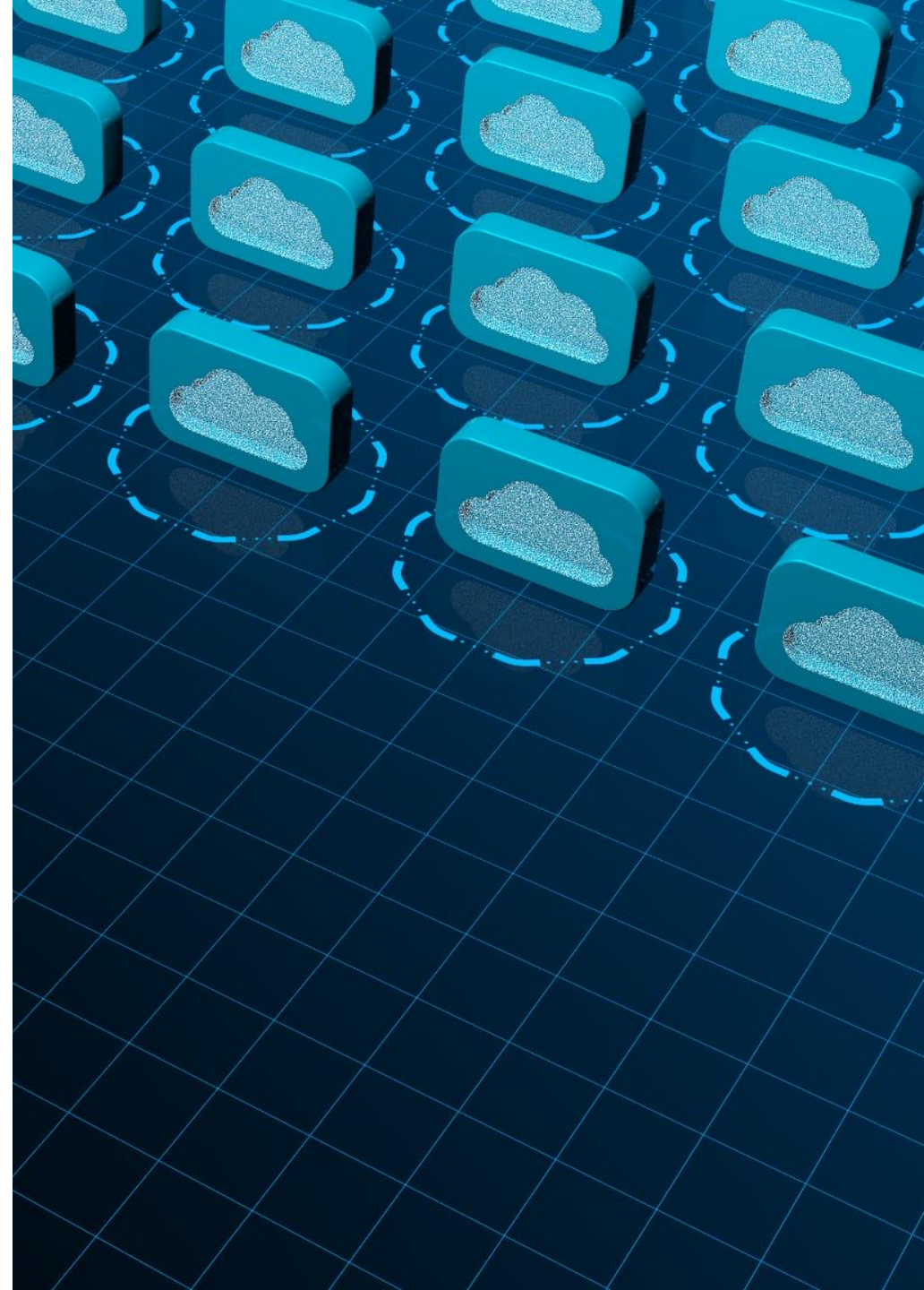
The CIDR range specifies the VPC's available IP addresses for subnets and network resources.

Impact on Scalability

Selecting the correct CIDR block is vital for future growth, allowing expansion and integration with other networks.

Avoiding Address Conflicts

Proper planning of CIDR blocks helps prevent address overlaps and ensures seamless network integration.



Understanding CIDR – IPv4

- Classless Inter-Domain Routing – a method for allocating IP addresses
- Used in Security Groups rules and AWS networking in general

IP version	Type	Protocol	Port range	Source	Description
IPv4	SSH	TCP	22	122.149.196.85/32	–
IPv4	HTTP	TCP	80	0.0.0.0/0	–

- They help to define an IP address range:
 - We've seen WW.XX.YY.ZZ/32 => one IP
 - We've seen 0.0.0.0/0 => all IPs
 - But we can define: 192.168.0.0/26 => 192.168.0.0 – 192.168.0.63 (64 IP addresses)

IPV4 IP addressing (CIDR)

0.0.0.0 ---- 255.255.255.255

XX.XX.XX.XX/26

XX – Octet – 8 bits

2 to the power 8 = 256

XX.XX.XX.XX/24

$32 - 26 = 6$

2 to the power 6 = 64

192.168.0.0/24 – 192.168.0.0 – 255.255.255.0

$32 - 24 = 8$

IP address EC2: 192.168.0.5

Netmask : 255.255.255.0

Understanding CIDR – IPv4

- A CIDR consists of two components
- Base IP
 - Represents an IP contained in the range (XX.XX.XX.XX)
 - Example: 10.0.0.0, 192.168.0.0, ...
- Subnet Mask
 - Defines how many bits can change in the IP
 - Example: /0, /24, /32
 - Can take two forms:
 - /8 $\Leftrightarrow$ 255.0.0.0
 - /16 $\Leftrightarrow$ 255.255.0.0
 - /24 $\Leftrightarrow$ 255.255.255.0
 - /32 $\Leftrightarrow$ 255.255.255.255

Understanding CIDR – Subnet Mask

- The Subnet Mask basically allows part of the underlying IP to get additional next values from the base IP

192	168	0	0	/32 => allows for 1 IP (2^0)	—————> 192.168.0.0
192	168	0	0	/31 => allows for 2 IP (2^1)	—————> 192.168.0.0 -> 192.168.0.1
192	168	0	0	/30 => allows for 4 IP (2^2)	—————> 192.168.0.0 -> 192.168.0.3
192	168	0	0	/29 => allows for 8 IP (2^3)	—————> 192.168.0.0 -> 192.168.0.7
192	168	0	0	/28 => allows for 16 IP (2^4)	—————> 192.168.0.0 -> 192.168.0.15
192	168	0	0	/27 => allows for 32 IP (2^5)	—————> 192.168.0.0 -> 192.168.0.31
192	168	0	0	/26 => allows for 64 IP (2^6)	—————> 192.168.0.0 -> 192.168.0.63
192	168	0	0	/25 => allows for 128 IP (2^7)	—————> 192.168.0.0 -> 192.168.0.127
192	168	0	0	/24 => allows for 256 IP (2^8)	—————> 192.168.0.0 -> 192.168.0.255
...					
192	168	0	0	/16 => allows for 65,536 IP (2^{16})	—————> 192.168.0.0 -> 192.168.255.255
...					
192	168	0	0	/0 => allows for All IPs	—————> 0.0.0.0 -> 255.255.255.255



Quick Memo

Octets

1st 2nd 3rd 4th

- **/32** – no octet can change
- **/24** – last octet can change
- **/16** – last 2 octets can change
- **/8** – last 3 octets can change
- **/0** – all octets can change

Understanding CIDR – Little Exercise

- 192.168.0.0/24 = ... ?
 - 192.168.0.0 – 192.168.0.255 (256 IPs)
- 192.168.0.0/16 = ... ?
 - 192.168.0.0 – 192.168.255.255 (65,536 IPs)
- 134.56.78.123/32 = ... ?
 - Just 134.56.78.123
- 0.0.0.0/0
 - All IPs!
- When in doubt, use this website <https://www.ipaddressguide.com/cidr>

Public vs. Private IP (IPv4)

- The Internet Assigned Numbers Authority (IANA) established certain blocks of IPv4 addresses for the use of private (LAN) and public (Internet) addresses
- Private IP can only allow certain values:
 - 10.0.0.0 – 10.255.255.255 (10.0.0.0/8) ← in big networks
 - 172.16.0.0 – 172.31.255.255 (172.16.0.0/12) ← AWS default VPC in that range
 - 192.168.0.0 – 192.168.255.255 (192.168.0.0/16) ← e.g., home networks
- All the rest of the IP addresses on the Internet are Public

Default VPC Walkthrough

- All new AWS accounts have a default VPC
- New EC2 instances are launched into the default VPC if no subnet is specified
- Default VPC has Internet connectivity and all EC2 instances inside it have public IPv4 addresses
- We also get a public and a private IPv4 DNS names



VPC in AWS – IPv4

- VPC = Virtual Private Cloud
- You can have multiple VPCs in an AWS region (max. 5 per region – soft limit)
- Max. CIDR per VPC is 5, for each CIDR:
 - Min. size is /28 (16 IP addresses)
 - Max. size is /16 (65536 IP addresses)
- Because VPC is private, only the Private IPv4 ranges are allowed:
 - 10.0.0.0 – 10.255.255.255 (10.0.0.0/8)
 - 172.16.0.0 – 172.31.255.255 (172.16.0.0/12)
 - 192.168.0.0 – 192.168.255.255 (192.168.0.0/16)
- Your VPC CIDR should NOT overlap with your other networks (e.g., corporate)



Manually Creating a VPC

Access VPC Dashboard

Start by logging into the cloud console and open the VPC dashboard to begin the creation process.

Configure VPC Settings

Set the IPv4 CIDR block and name the VPC. These are essential for network setup and identification.

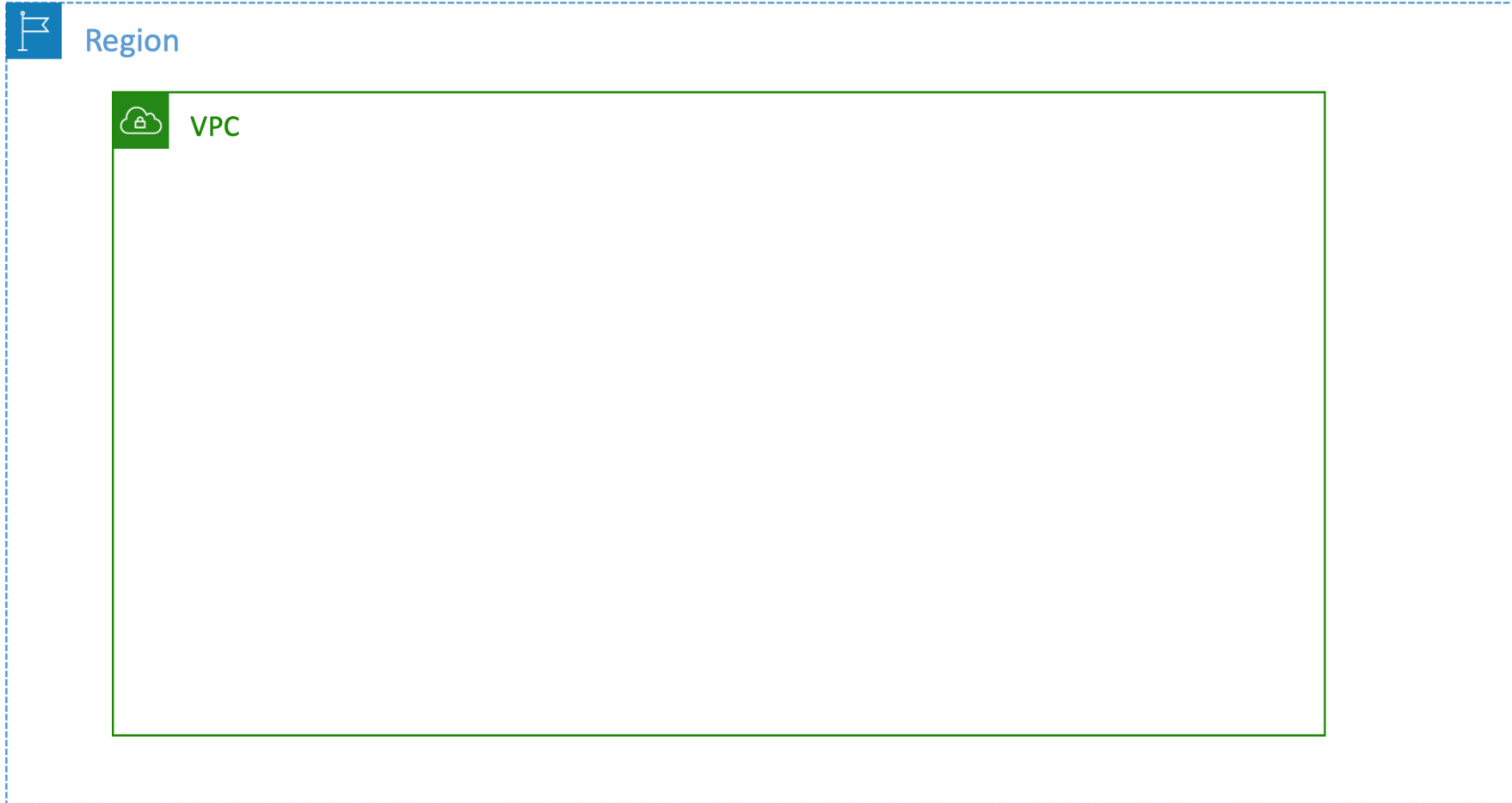
Set Up Subnets and Connectivity

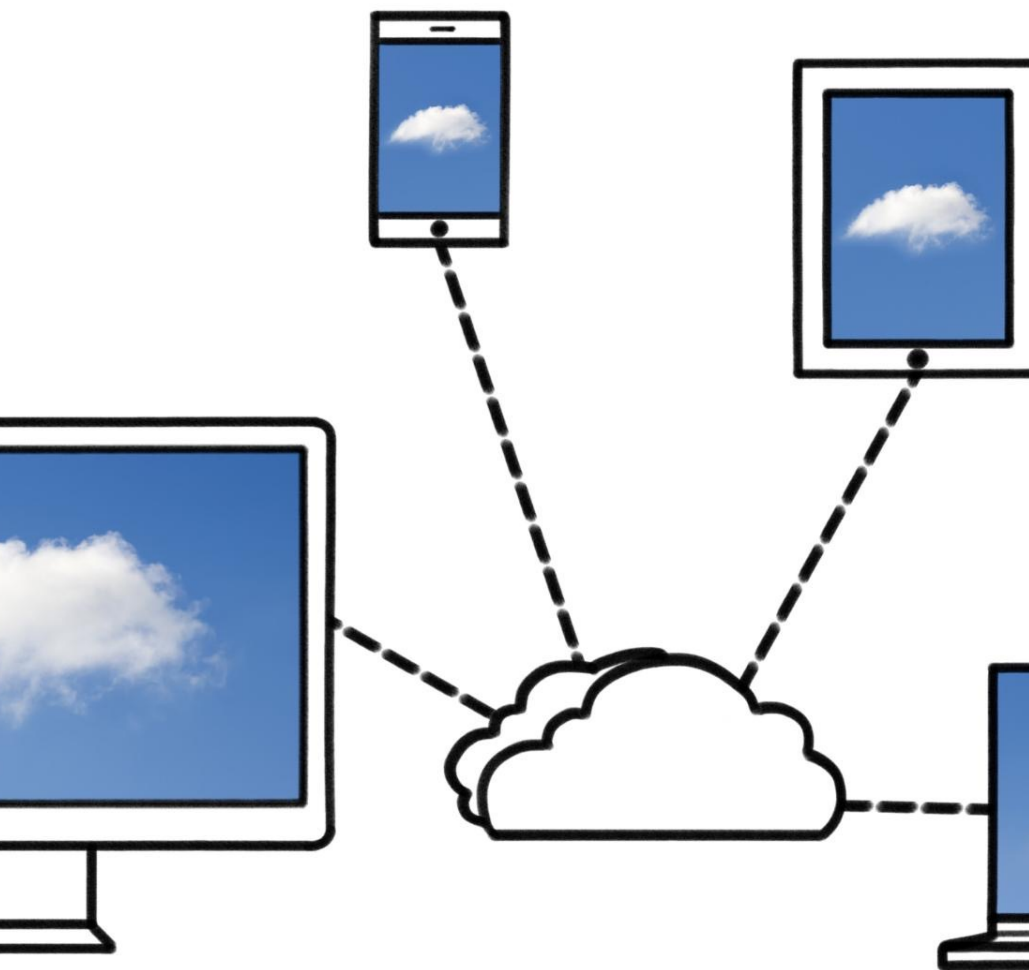
Add subnets in multiple availability zones and configure an Internet Gateway for external access.

Update Route Tables

Modify route tables to direct traffic for subnets, enabling external connectivity as needed.

State of Hands-on





Understanding Default Subnets in AWS

Automatic Creation in Zones

Default subnets are provisioned automatically in each Availability Zone with a default VPC, simplifying network setup.

Immediate Resource Deployment

Resources can be deployed instantly in default subnets without additional configuration, speeding up the process for users.

Public Internet Accessibility

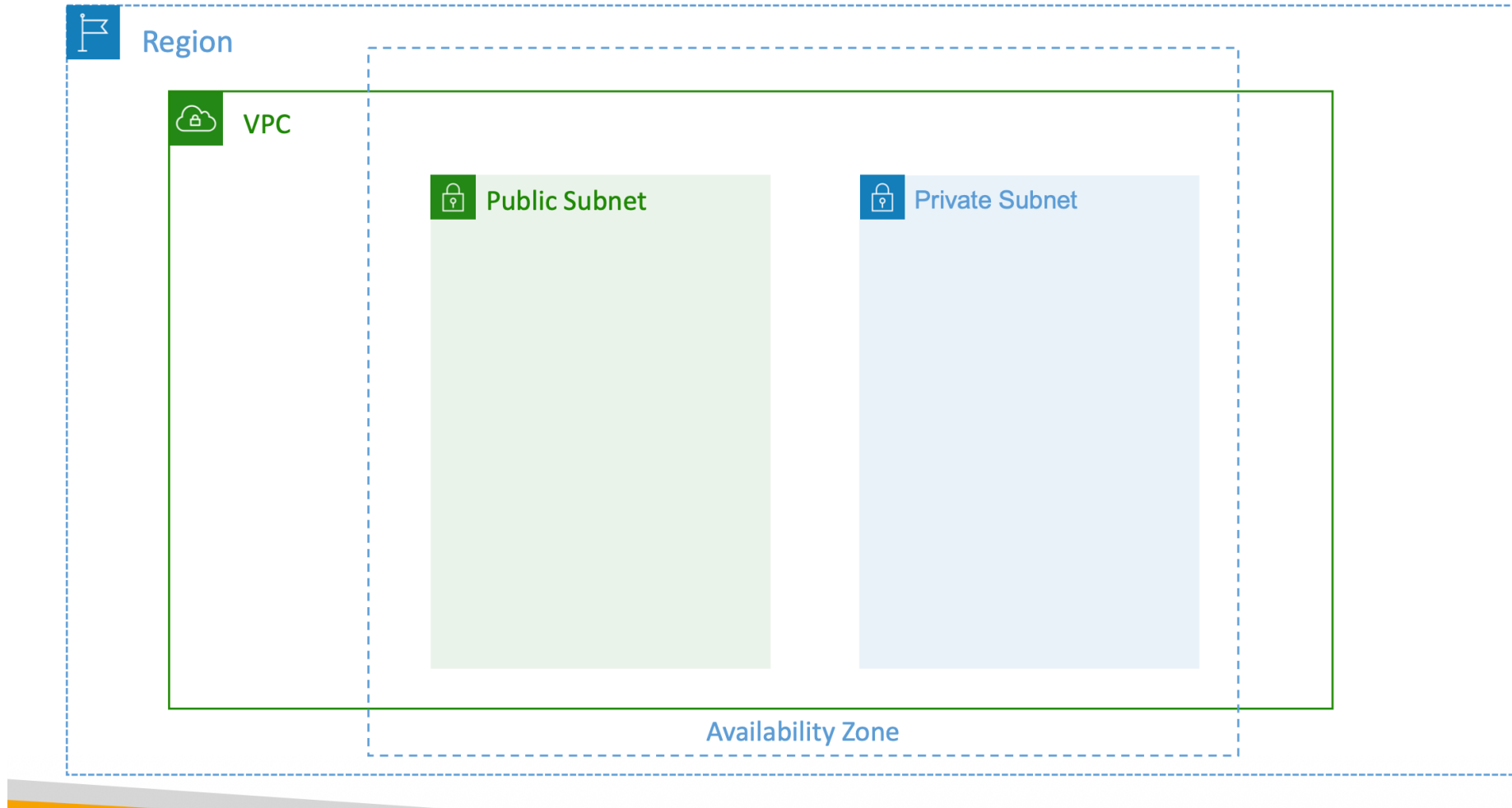
Default subnets allow public IP assignment, enabling resources to be accessed from the internet and easing connectivity.

VPC – Subnet (IPv4)



- AWS reserves 5 IP addresses (first 4 & last 1) in each subnet
- These 5 IP addresses are not available for use and can't be assigned to an EC2 instance
- Example: if CIDR block 10.0.0.0/24, then reserved IP addresses are:
 - 10.0.0.0 – Network Address
 - 10.0.0.1 – reserved by AWS for the VPC router
 - 10.0.0.2 – reserved by AWS for mapping to Amazon-provided DNS
 - 10.0.0.3 – reserved by AWS for future use
 - 10.0.0.255 – Network Broadcast Address. AWS does not support broadcast in a VPC, therefore the address is reserved
- Exam Tip, if you need 29 IP addresses for EC2 instances:
 - You can't choose a subnet of size /27 (32 IP addresses, $32 - 5 = 27 < 29$)
 - You need to choose a subnet of size /26 (64 IP addresses, $64 - 5 = 59 > 29$)

Adding Subnets





Setting Up an Internet Gateway

Purpose of Internet Gateway

An Internet Gateway provides connectivity from AWS VPC resources to the internet, vital for public applications.

Creating the Gateway

On the VPC dashboard, select 'Internet Gateways,' then create and name a new gateway for your setup.

Attaching to VPC

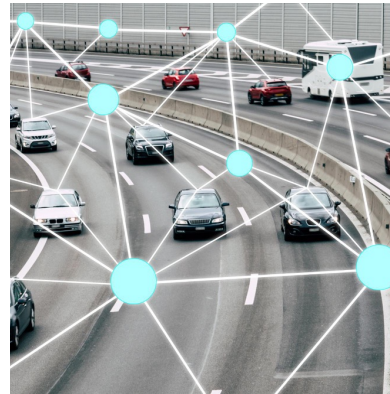
After creation, attach the Internet Gateway to your VPC to enable internet access for the connected resources.

Understanding AWS VPC Route Tables



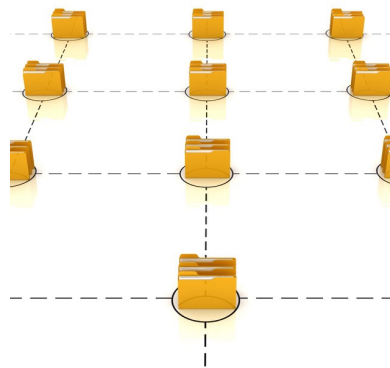
Role of Route Tables

Route tables manage the direction of network packets within AWS VPC, controlling how traffic flows in the cloud.



Subnet Association

Each subnet in a VPC needs to be linked to a route table to ensure correct traffic routing and connectivity.



Routing to Destinations

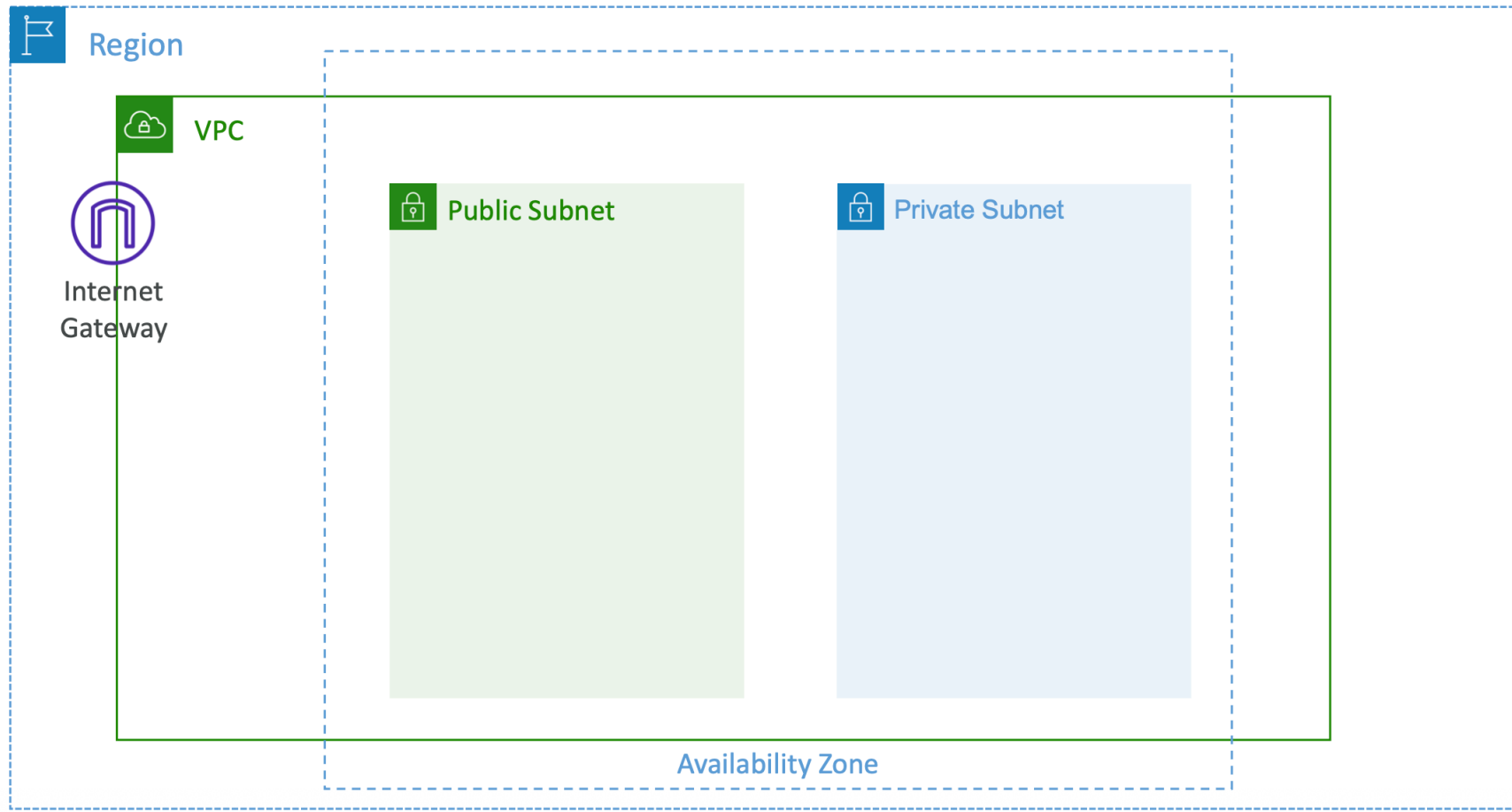
Route tables handle traffic between subnets, internet gateways, and other destinations, enabling secure network communication.

Internet Gateway (IGW)

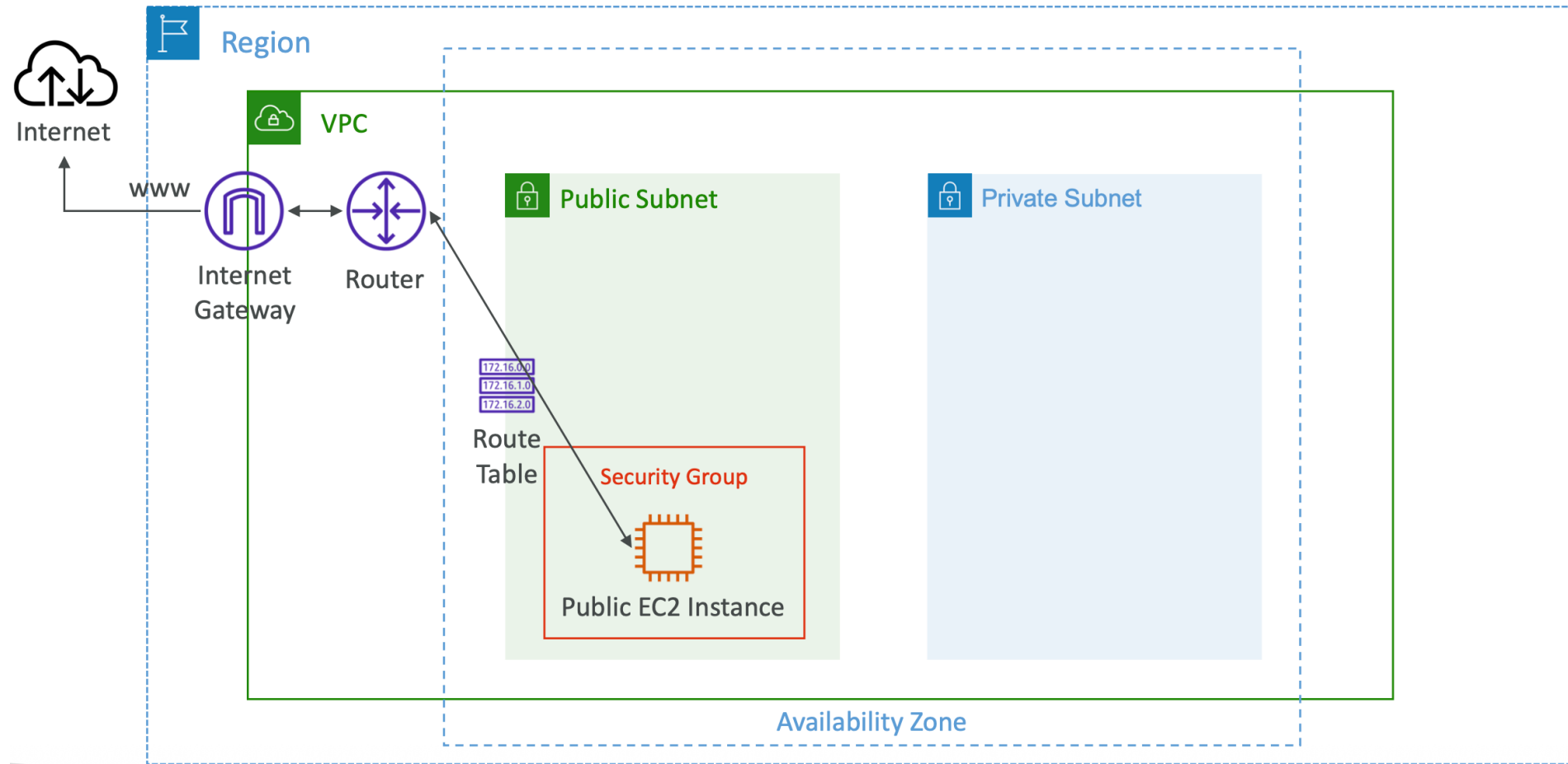


- Allows resources (e.g., EC2 instances) in a VPC connect to the Internet
 - It scales horizontally and is highly available and redundant
 - Must be created separately from a VPC
 - One VPC can only be attached to one IGW and vice versa
-
- Internet Gateways on their own do not allow Internet access...
 - Route tables must also be edited!

Adding Internet Gateway

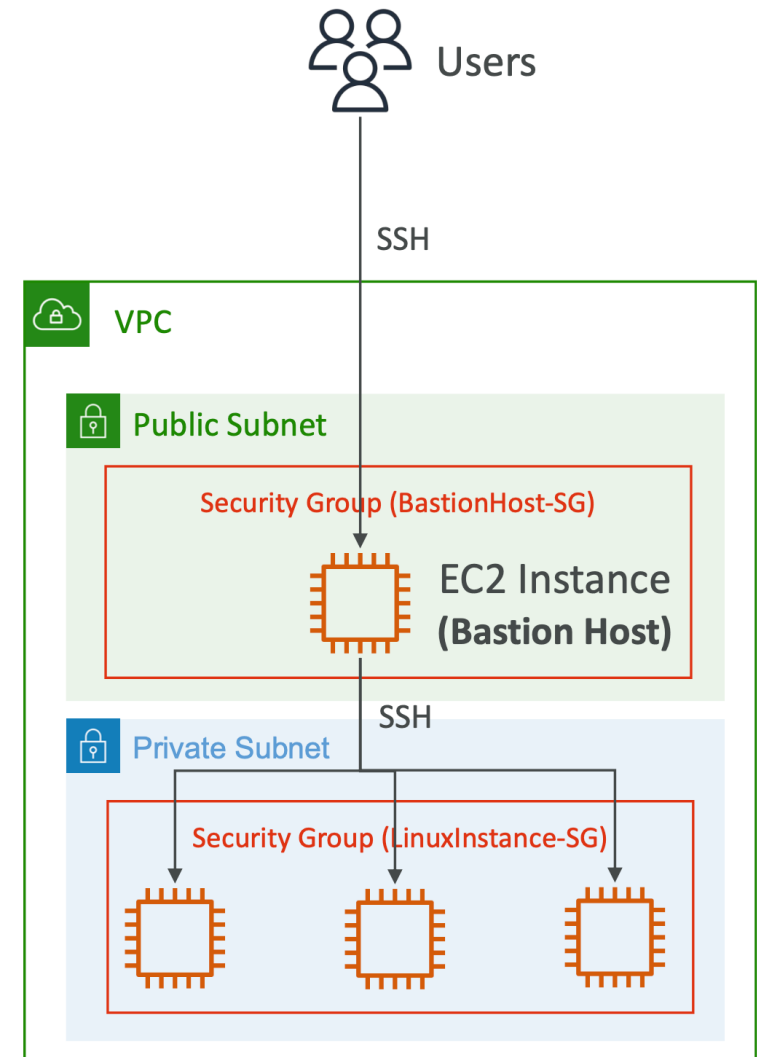


Editing Route Tables



Bastion Hosts

- We can use a Bastion Host to SSH into our private EC2 instances
- The bastion is in the public subnet which is then connected to all other private subnets
- Bastion Host security group must allow inbound from the internet on port 22 from restricted CIDR, for example the public CIDR of your corporation
- Security Group of the EC2 Instances must allow the Security Group of the Bastion Host, or the private IP of the Bastion host

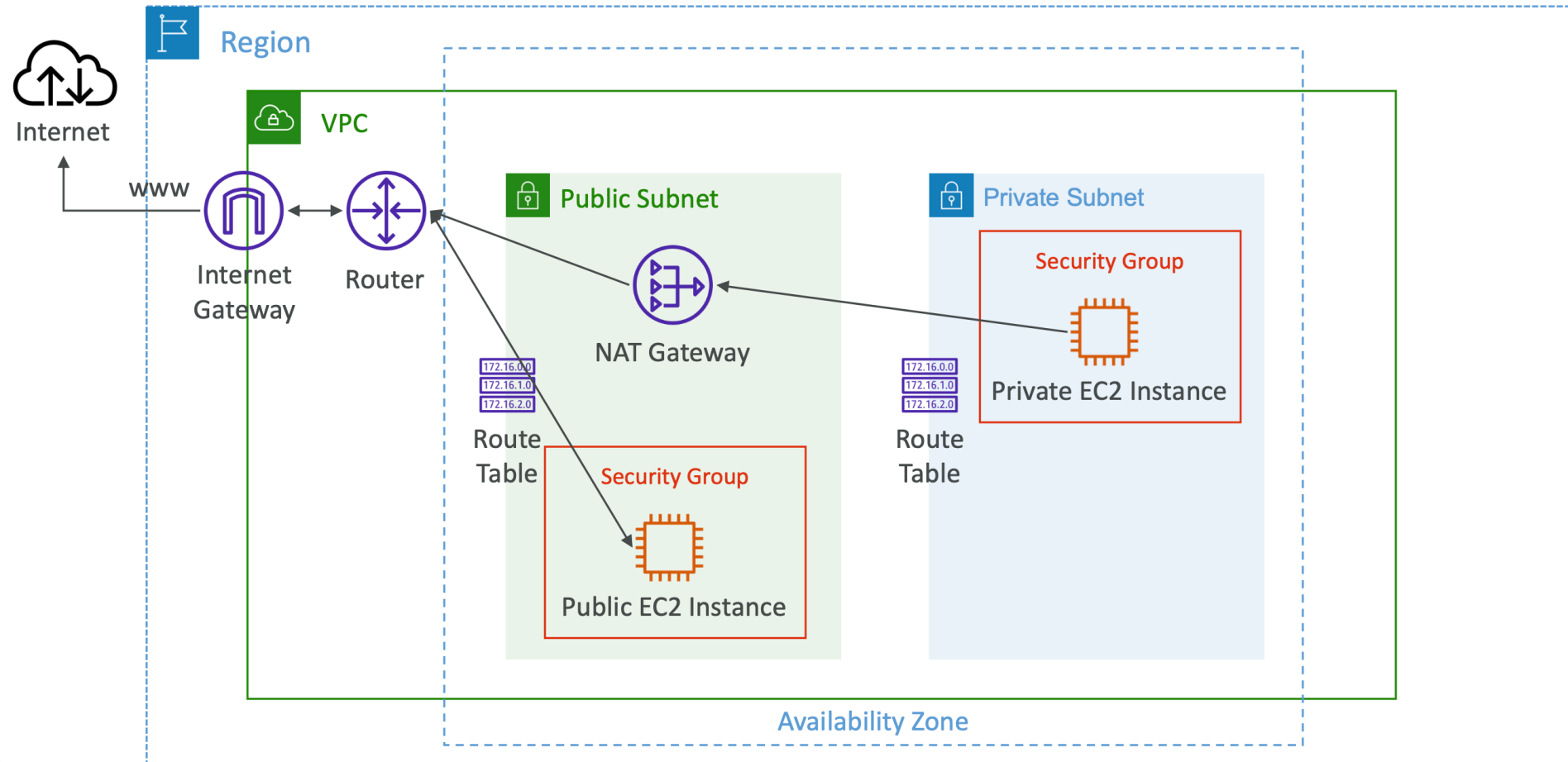


NAT Gateway



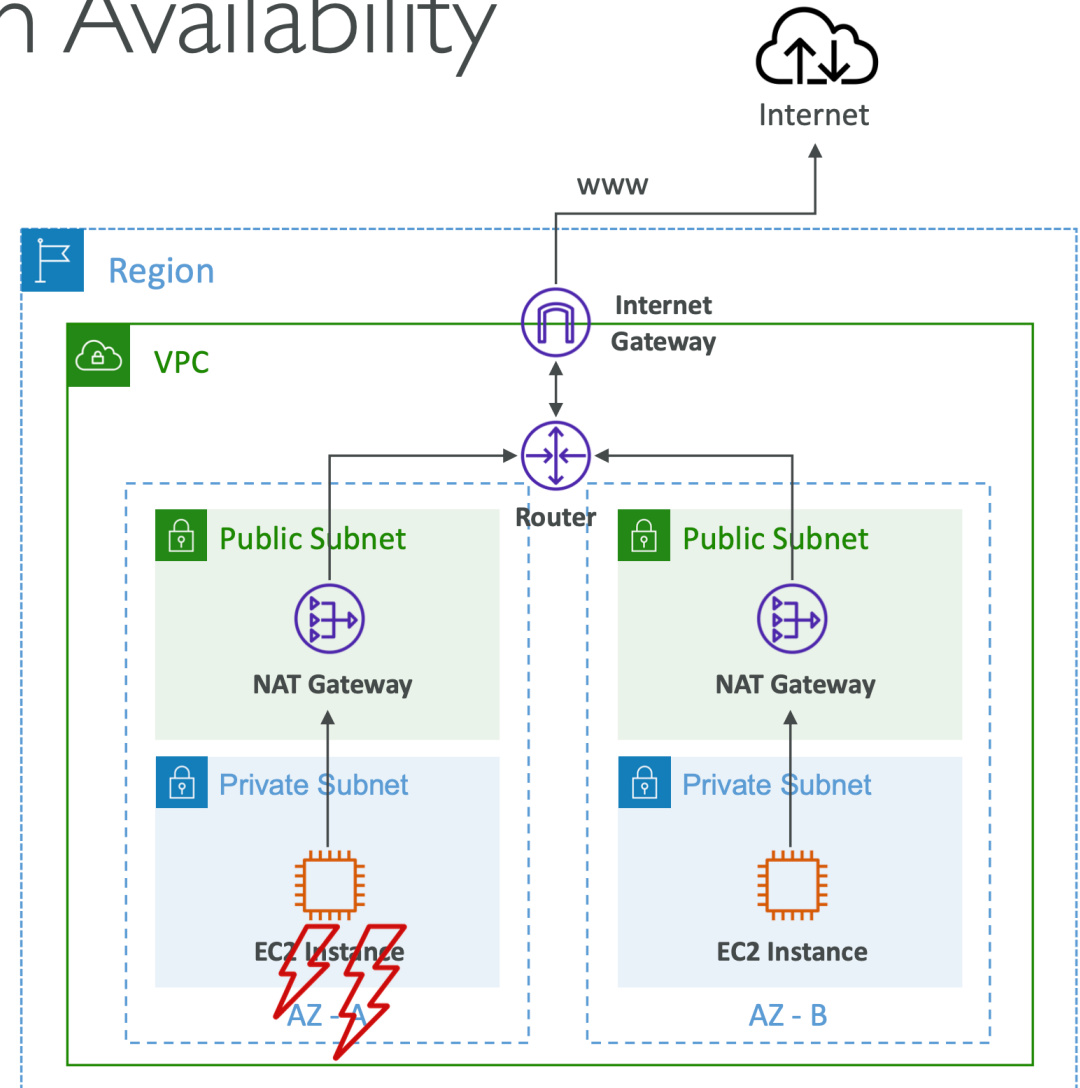
- AWS-managed NAT, higher bandwidth, high availability, no administration
- Pay per hour for usage and bandwidth
- NATGW is created in a specific Availability Zone, uses an Elastic IP
- Can't be used by EC2 instance in the same subnet (only from other subnets)
- Requires an IGW (Private Subnet => NATGW => IGW)
- 5 Gbps of bandwidth with automatic scaling up to 100 Gbps
- No Security Groups to manage / required

NAT Gateway



NAT Gateway with High Availability

- NAT Gateway is resilient within a single Availability Zone
- Must create multiple NAT Gateways in multiple AZs for fault-tolerance
- There is no cross-AZ failover needed because if an AZ goes down it doesn't need NAT





AWS VPC Network ACLs

Subnet-Level Security

Network ACLs offer an extra security layer at the subnet level, managing inbound and outbound traffic effectively.

Stateless Rule Management

NACLs are stateless and allow detailed rules for IP addresses and protocols, enabling flexible traffic management.

Compliance and Segmentation

NACLs help secure sensitive apps, meet compliance, and efficiently segment network traffic in real-time environments.

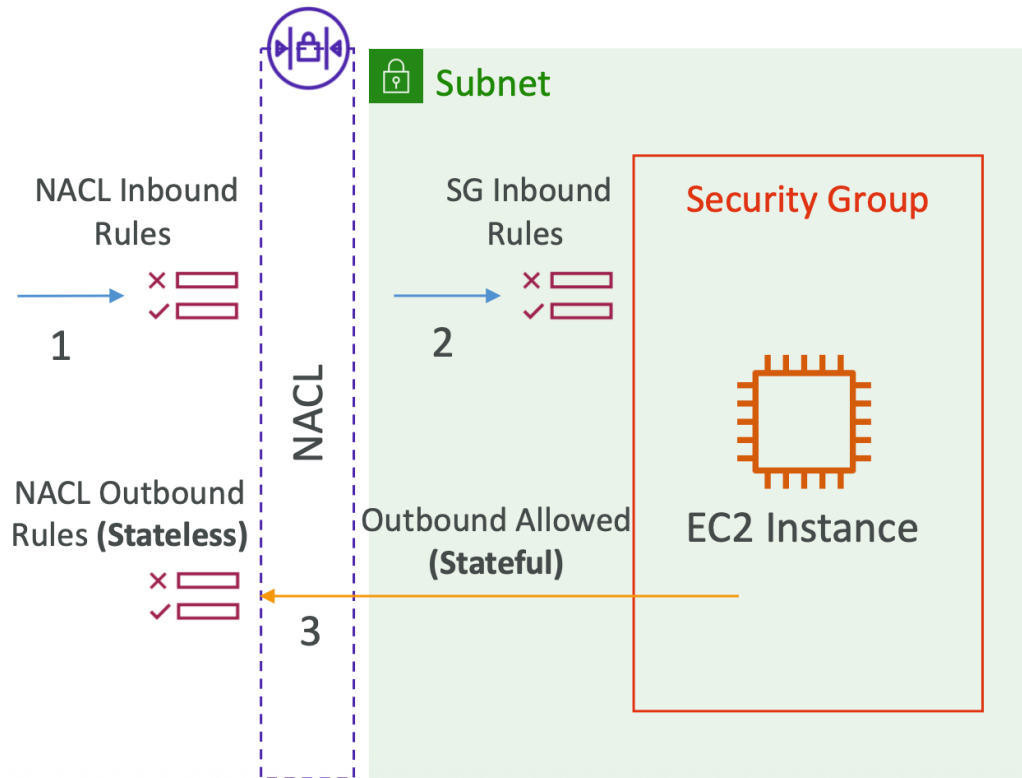


Network Access Control List (NACL)

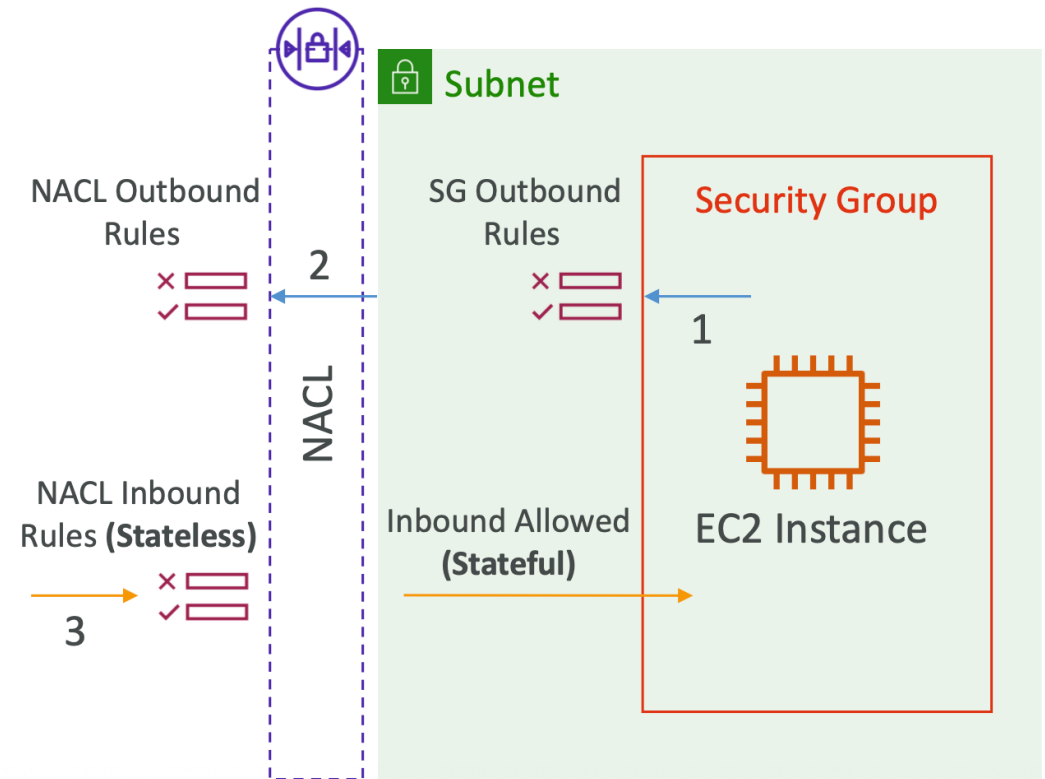
- NACL are like a firewall which control traffic from and to subnets
- One NACL per subnet, new subnets are assigned the Default NACL
- You define NACL Rules:
 - Rules have a number (1-32766), higher precedence with a lower number
 - First rule match will drive the decision
 - Example: if you define #100 ALLOW 10.0.0.10/32 and #200 DENY 10.0.0.10/32, the IP address will be allowed because 100 has a higher precedence over 200
 - The last rule is an asterisk (*) and denies a request in case of no rule match
 - AWS recommends adding rules by increment of 100
- Newly created NACLs will deny everything
- NACL are a great way of blocking a specific IP address at the subnet level

Security Groups & NACLs

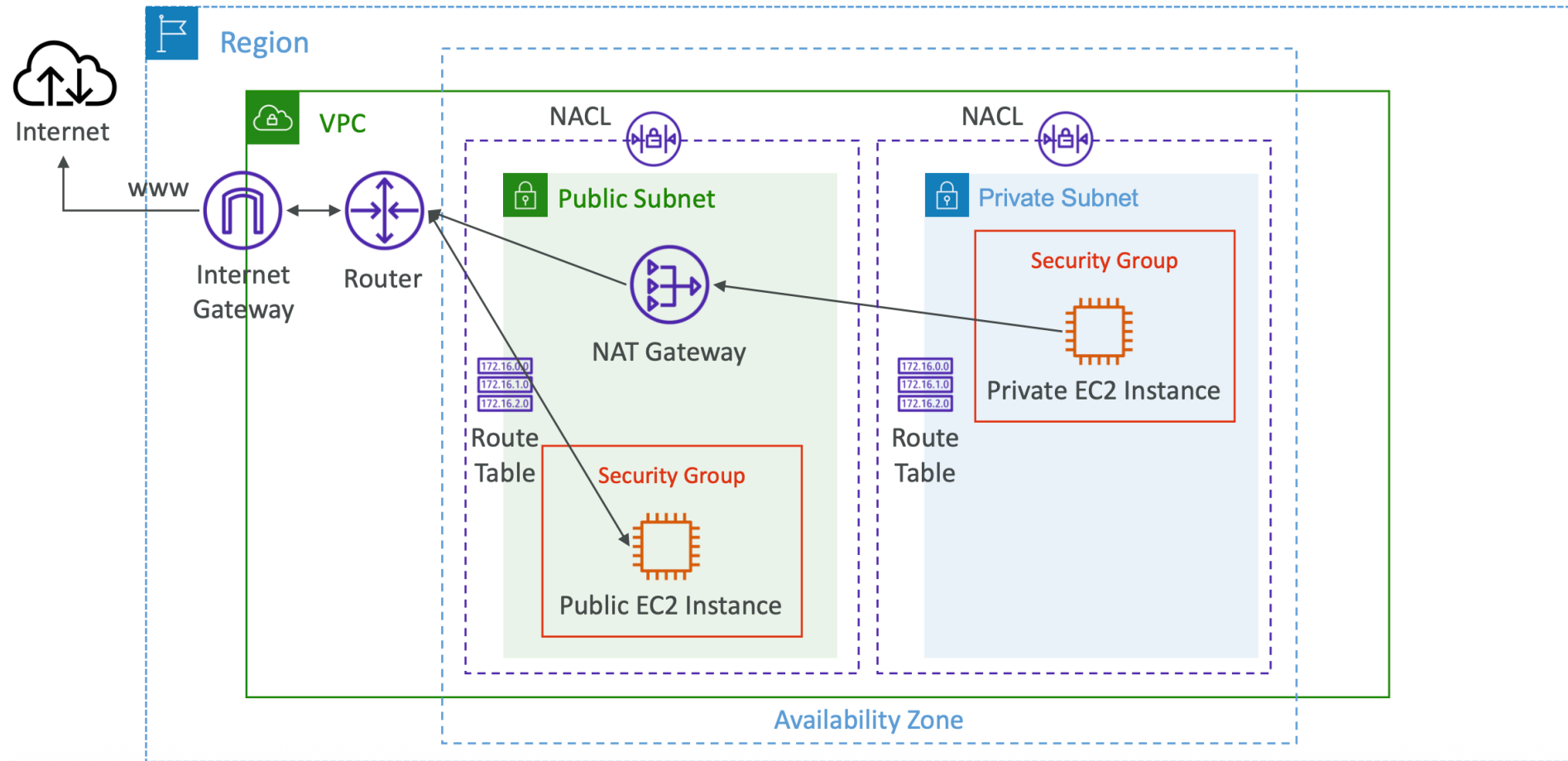
Incoming Request



Outgoing Request



NACLs



Default NACL

- Accepts everything inbound/outbound with the subnets it's associated with
- Do NOT modify the Default NACL, instead create custom NACLs



Default NACL for a VPC that supports IPv4

Inbound Rules

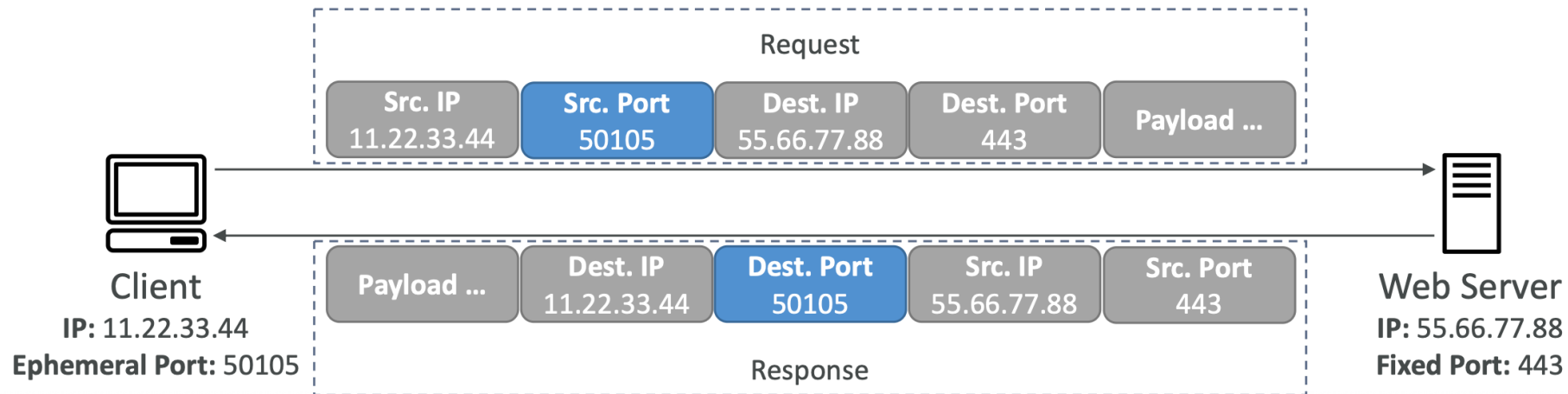
Rule #	Type	Protocol	Port Range	Source	Allow/Deny
100	All IPv4 Traffic	All	All	0.0.0.0/0	ALLOW
*	All IPv4 Traffic	All	All	0.0.0.0/0	DENY

Outbound Rules

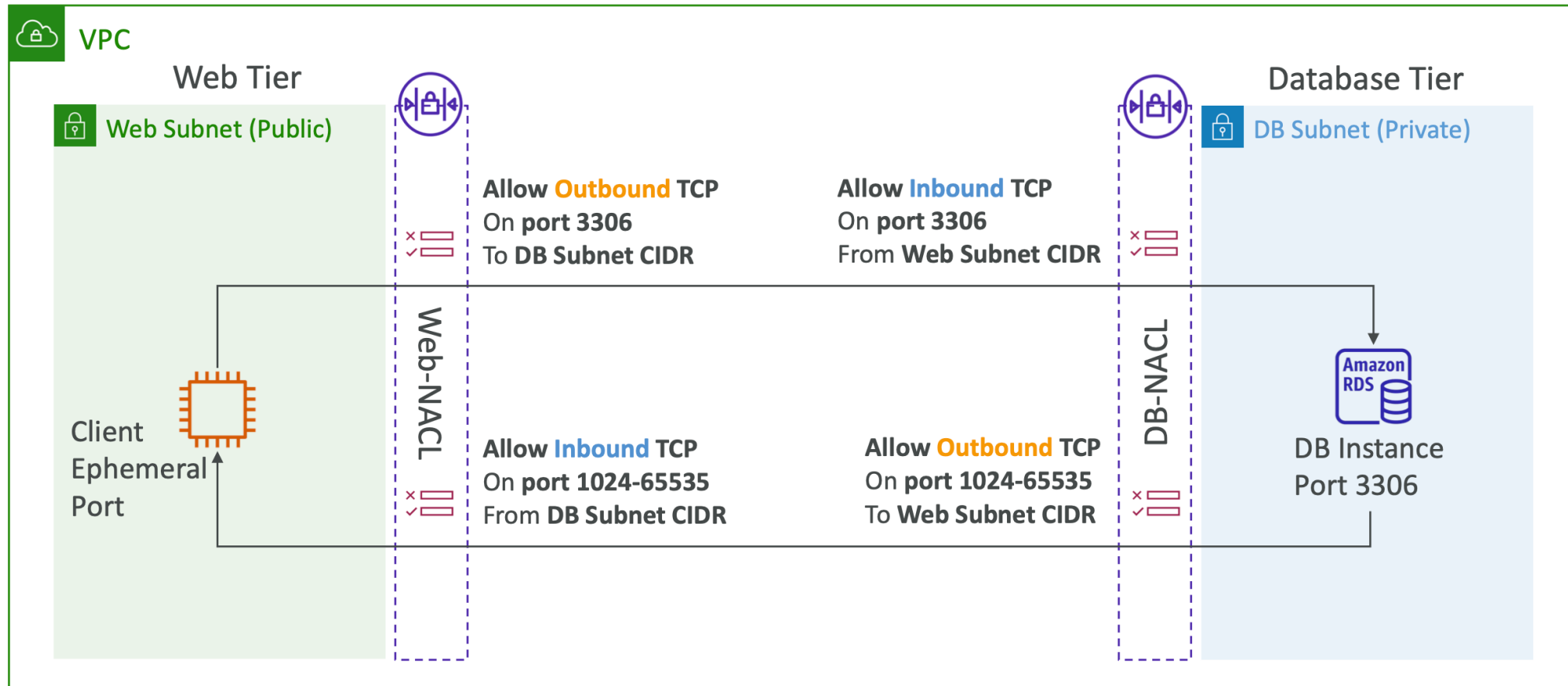
Rule #	Type	Protocol	Port Range	Destination	Allow/Deny
100	All IPv4 Traffic	All	All	0.0.0.0/0	ALLOW
*	All IPv4 Traffic	All	All	0.0.0.0/0	DENY

Ephemeral Ports

- For any two endpoints to establish a connection, they must use ports
- Clients connect to a **defined port**, and expect a response on an **ephemeral port**
- Different Operating Systems use different port ranges, examples:
 - IANA & MS Windows 10 → 49152 – 65535
 - Many Linux Kernels → 32768 – 60999



NACL with Ephemeral Ports



<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-network-acls.html#nacl-ephemeral-ports>

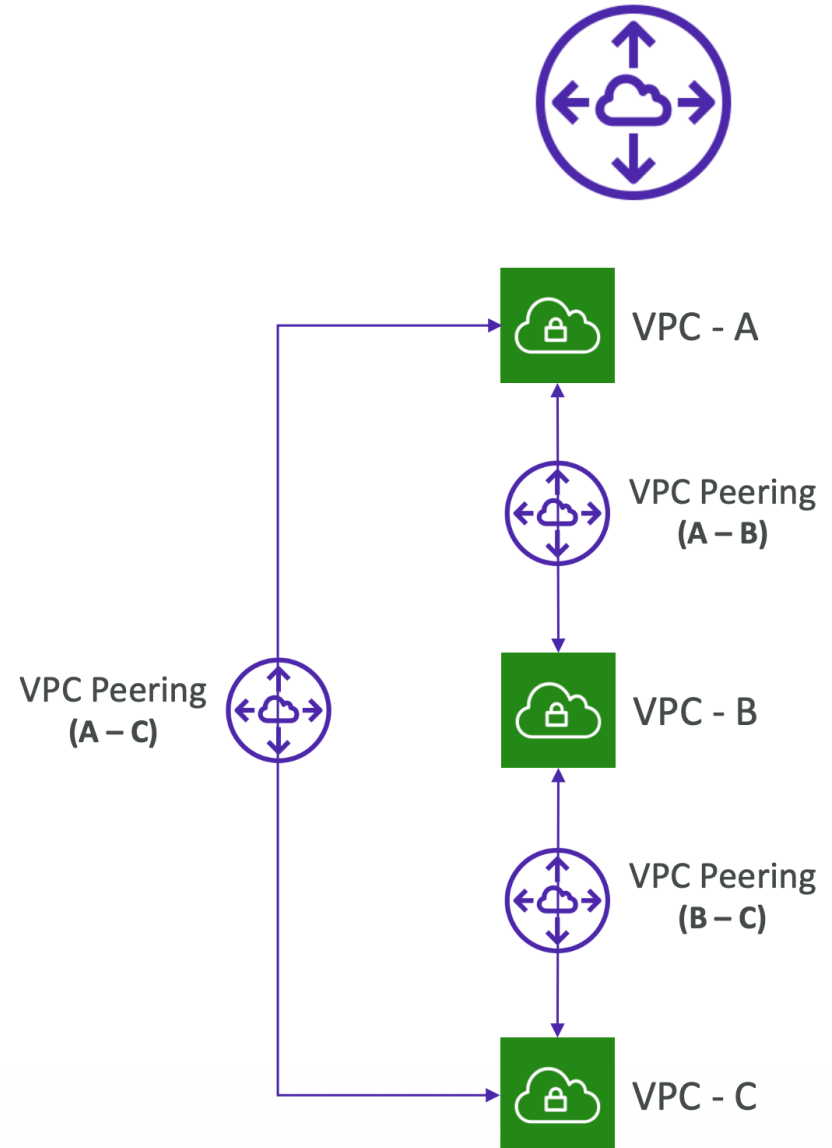
Security Group vs. NACLs

Security Group	NACL
Operates at the instance level	Operates at the subnet level
Supports allow rules only	Supports allow rules and deny rules
Stateful: return traffic is automatically allowed, regardless of any rules	Stateless: return traffic must be explicitly allowed by rules (think of ephemeral ports)
All rules are evaluated before deciding whether to allow traffic	Rules are evaluated in order (lowest to highest) when deciding whether to allow traffic, first match wins
Applies to an EC2 instance when specified by someone	Automatically applies to all EC2 instances in the subnet that it's associated with

NACL Examples: <https://docs.aws.amazon.com/vpc/latest/userguide/vpc-network-acls.html>

VPC Peering

- Privately connect two VPCs using AWS' network
- Make them behave as if they were in the same network
- Must not have overlapping CIDRs
- VPC Peering connection is **NOT** transitive (must be established for each VPC that need to communicate with one another)
- You must update route tables in each VPC's subnets to ensure EC2 instances can communicate with each other



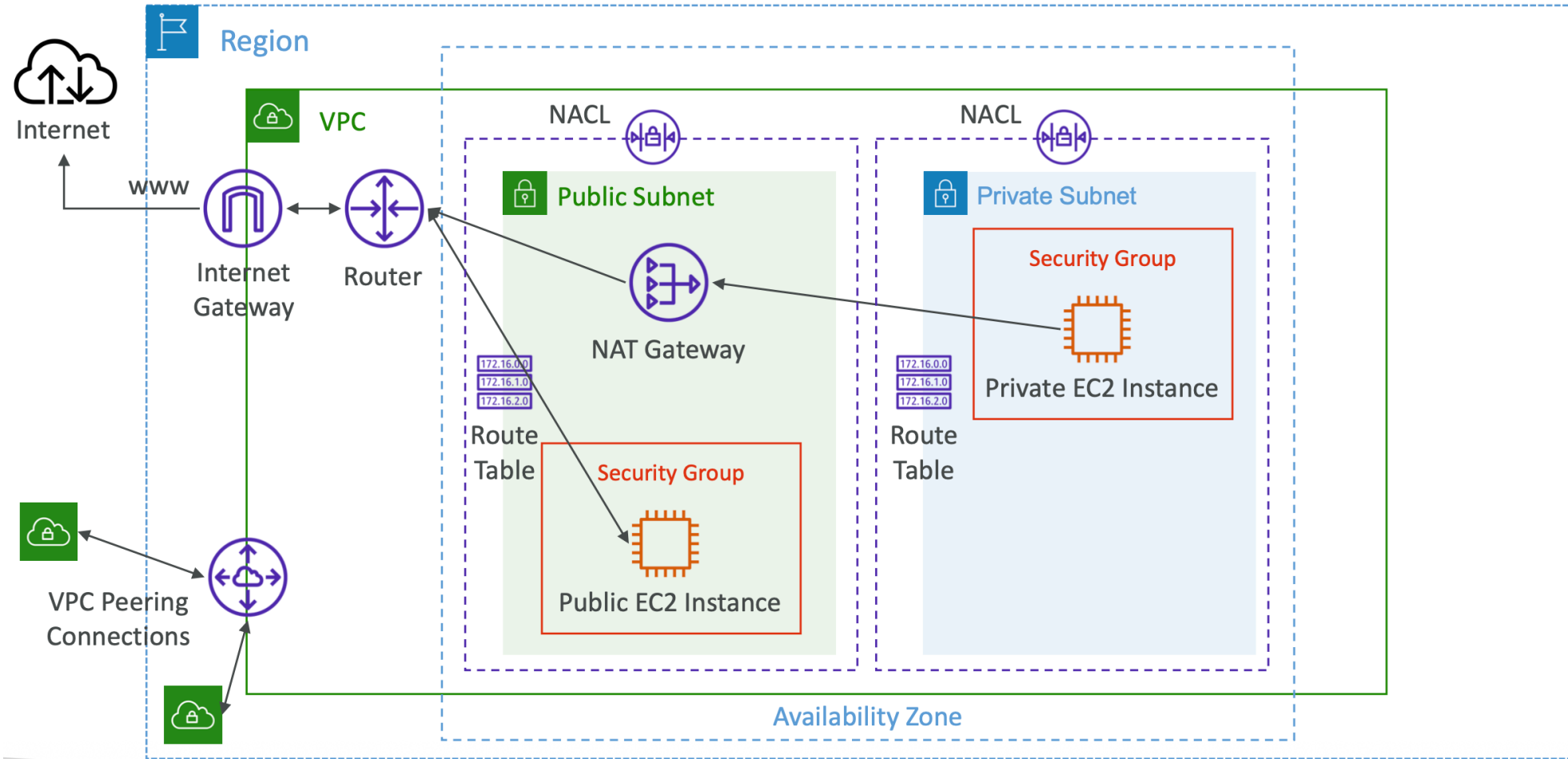
VPC Peering – Good to know

- You can create VPC Peering connection between VPCs in different AWS accounts/regions
- You can reference a security group in a peered VPC (works cross accounts – same region)

Type ▾	Protocol ▾	Port range ▾	Source ▾
HTTP	TCP	80	sg-04991f9af3473b939 / default
HTTP	TCP	80	/ sg-027ad1f7865d4be76

↑
Account ID

VPC Peering

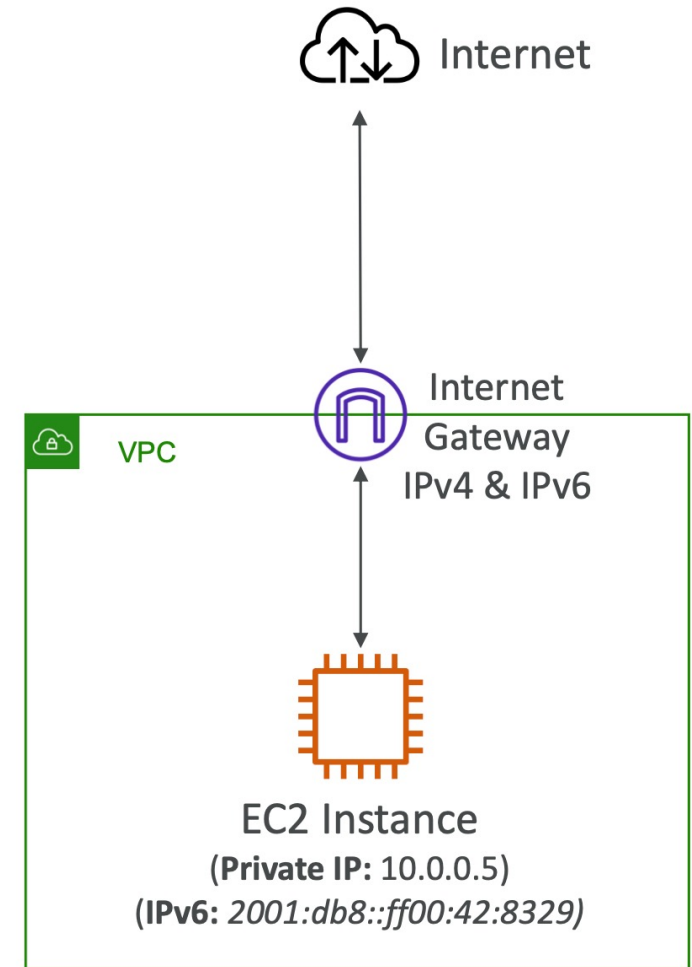


What is IPv6?

- IPv4 designed to provide 4.3 Billion addresses (they'll be exhausted soon)
- IPv6 is the successor of IPv4
- IPv6 is designed to provide 3.4×10^{38} unique IP addresses
- Every IPv6 address in AWS is public and Internet-routable (no private range)
- Format → x.x.x.x.x.x.x.x (x is hexadecimal, range can be from 0000 to ffff)
- Examples:
 - 2001:db8:3333:4444:5555:6666:7777:8888
 - 2001:db8:3333:4444:cccc:dddd:eeee:ffff
 - :: → all 8 segments are zero
 - 2001:db8:: → the last 6 segments are zero
 - ::1234:5678 → the first 6 segments are zero
 - 2001:db8::1234:5678 → the middle 4 segments are zero

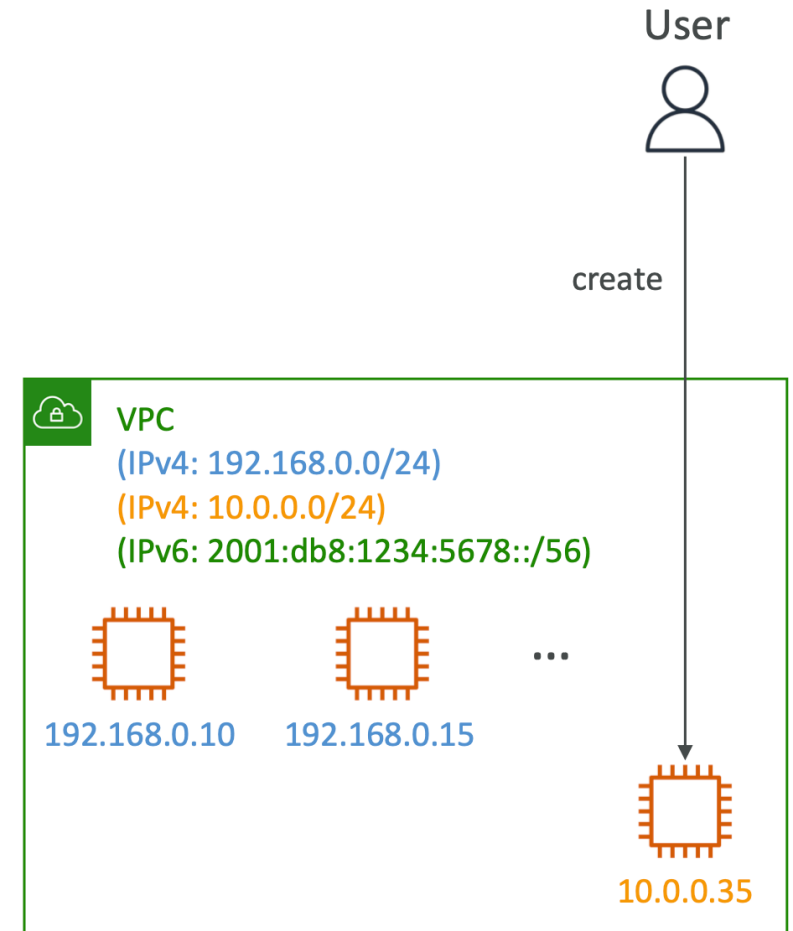
IPv6 in VPC

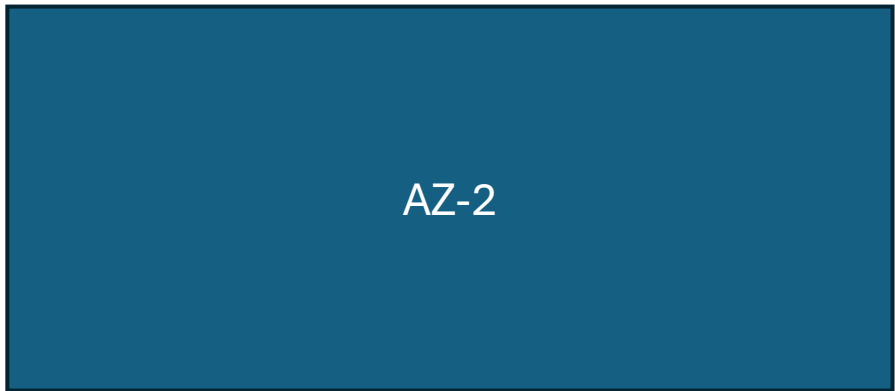
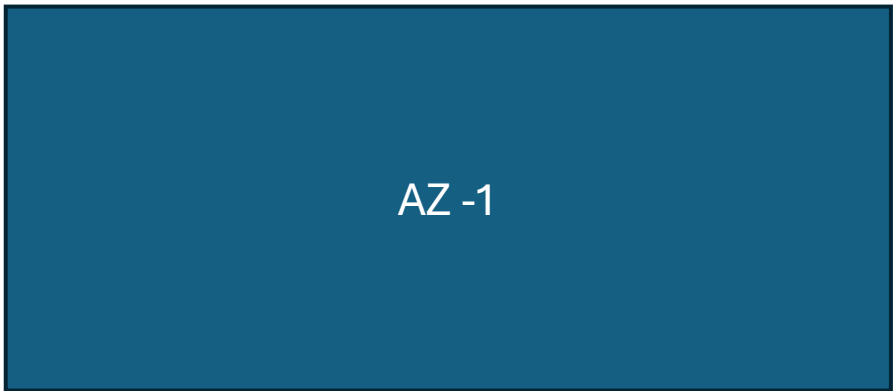
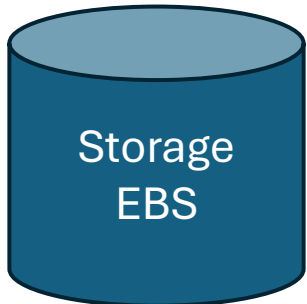
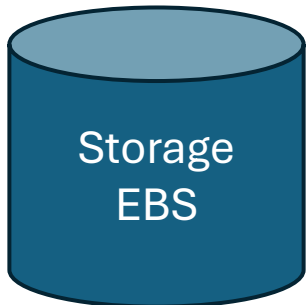
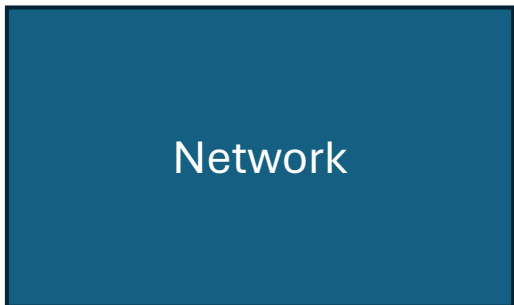
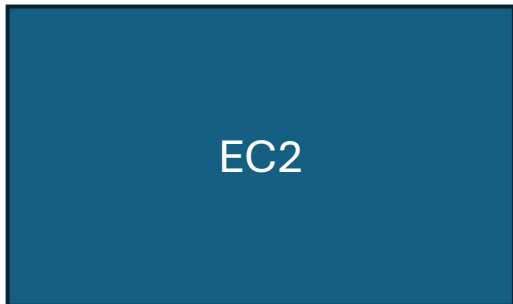
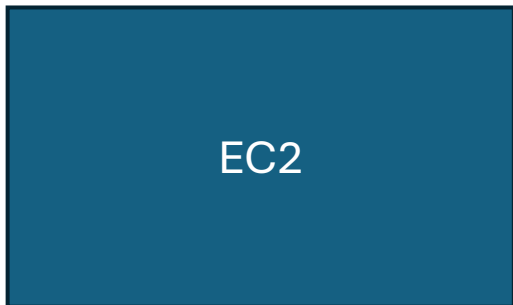
- IPv4 cannot be disabled for your VPC and subnets
- You can enable IPv6 (they're public IP addresses) to operate in dual-stack mode
- Your EC2 instances will get at least a private internal IPv4 and a public IPv6
- They can communicate using either IPv4 or IPv6 to the internet through an Internet Gateway



IPv4 Troubleshooting

- IPv4 cannot be disabled for your VPC and subnets
- So, if you cannot launch an EC2 instance in your subnet
 - It's not because it cannot acquire an IPv6 (the space is very large)
 - It's because there are no available IPv4 in your subnet
- Solution: create a new IPv4 CIDR in your subnet





App 1 → [Https://IP](#) address

App11 --> <https://xyz.com>

Xyz.com ↔ IP address – DNS server – port 53

https – 443

http – 80

Dns - 53

App2 → [Https://IP](#) address

[172.31.0.0](#) -→ [172.31.0.1](#)
[172.31.0.2](#)

[172.31.1.0](#). → 255

[172.31.2.0](#) -- 2555

[172.31.255.0](#)